

DRY-RUN GUIDE | 14_grade.cpp
Source: cpp\1st_belt\14_grade.cpp

CORE BEHAVIOR

Rejects scores outside 0..100; otherwise chooses the first applicable grade threshold from F through A.

REPRESENTATIVE INPUT / CALL

Input: '81'.

STEP TRACE

Valid range. <=39? no; <=50? no; <=60? no; <=70? no; <=80? no; else A.

EXPECTED OUTPUT / RETURN

'A'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | You are provided with the following table. Based on the student's score,
004 | print the appropriate grade.
005 |
006 | Grading Metrics:
007 | 0-39: F
008 | 40-50: E
009 | 51-60: D
010 | 61-70: C
011 | 71-80: B
012 | 81-100: A
013 |
014 | If an invalid score is provided outside the allowed range, print
015 | "Invalid Input".
016 | */
017 |
018 | #include <iostream>
019 | using namespace std;
020 |
021 | int main() {
022 |     int score;
023 |     cin >> score;
024 |
025 |     if (score < 0 || score > 100) {
026 |         cout << "Invalid Input";
027 |     } else if (score <= 39) {
028 |         cout << "F";
029 |     } else if (score <= 50) {
030 |         cout << "E";
031 |     } else if (score <= 60) {
032 |         cout << "D";
033 |     } else if (score <= 70) {
034 |         cout << "C";
035 |     } else if (score <= 80) {
036 |         cout << "B";
037 |     } else {
```

```
038 |         cout << "A";
039 |     }
040 |
041 |     return 0;
042 | }
043 |
```